(FP2) 3. Differentiation 作业2

NO. 1

Find the Maclaurin's series for $\tan(x + \frac{1}{4}\pi)$ up to and including the term in x^2. [5]

NO. 2

a. Starting from the definitions of tanh and sech in terms of exponentials, prove that $1 - \tanh^2\theta = \operatorname{sech}^2\theta$. [3]

The variables x and y are such that $\tanh y = \cos\left(x + \frac{1}{4}\pi\right)$, for $-\frac{1}{4}\pi < x < \frac{3}{4}\pi$.

b. By differentiating the equation $\tanh y = \cos\left(x + \frac{1}{4}\pi\right)$ with respect to x, show that $\frac{dy}{dx} = -\operatorname{cosec}\left(x + \frac{1}{4}\pi\right)$. [4]

c. Hence find the first three terms in the Maclaurin's series for $\tanh^{-1}\left(\cos\left(x + \frac{1}{4}\pi\right)\right)$ in the form $\frac{1}{2}\ln a + bx + cx^2$, giving the exact values of the constants a, b and c. [5]

NO. 3

It is given that $y = 2^x$.

a. By differentiating $\ln y$ with respect to x, show that $\frac{dy}{dx} = 2^x \ln 2$. [3]

b. Write down $\frac{d^2y}{dx^2}$. [1]

c. Hence find the first three terms in the Maclaurin's series for 2^x. [3]

NO. 4

Find the Maclaurin's series for $e^{1+x^2} + e^{1-x}$ up to and including the term in x^2 . [4]

NO. 5

It is given that $y = \sinh(x^2) + \cosh(x^2)$.

a. Use standard results from the list of formulae (MF19) to find the Maclaurin's series for y in terms of x up to and including the term in x^4. [2]

b. Deduce the value of $\frac{d^4y}{dx^4}$ when $x = 0$. [1]

c. Use your answer to part **(a)** to find an approximation to $\int_0^{\frac{1}{2}} y \, dx$, giving your answer as a rational fraction in its lowest terms. [2]

NO. 6

Find the Maclaurin's series for $e^{x} \tan x$ from first principles up to and including the term in x^2. [5]

NO. 7

a. It is given that $y = \text{sech}^{-1}(x+\frac{1}{2})$.

Express cosh y in terms of x and hence show that $\sinh y \frac{dy}{dx} = -\frac{1}{\left(x+\frac{1}{2}\right)^2}$. [3]

b. Find the first three terms in the Maclaurin's series for $\text{sech}^{-1}(x+\frac{1}{2})$ in the form

$\ln a + bx + cx^2$,

where a, b and c are constants to be determined. [7]